10 Lecture 10 (February 5, 2025): Reconstruction of an affine group

Scribe: Jackson Morris

We prove the reconstruction statement announced in Lecture 9.

10.1 Theorem. *For an affine group scheme G over a field k, the natural morphism*

$$G \longrightarrow \underline{\mathrm{Aut}}^{\otimes}(\omega), \qquad \omega : \mathrm{Rep}_k(G) \to \mathrm{Vec}_k,$$

is an isomorphism of group-valued functors. In particular, the tensor automorphism functor is representable.

Chevalley's theorem

10.2 Theorem (Chevalley)**.** *Every closed subgroup scheme H of an affine algebraic group K is the stabilizer of a one-dimensional subspace in a finite-dimensional representation of K.*

See [Mil12, Chapter VIII, Theorem 13.1].

Proof of Theorem 10.1. Fix $X \in \mathrm{Rep}_k(G)$ and put $H = G_X$ and $K = \mathrm{GL}(X)$. As in Lecture 9, the tensor subcategory $\mathcal{C}_X$ generated by X identifies with $\mathrm{Rep}(H)$, and for every commutative k-algebra R we have inclusions

$$H(R) \subseteq \underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X})(R) \subseteq K(R).$$

Let λ be a tensor automorphism in the middle group and write $g = \lambda_X \in K(R)$. By Chevalley's theorem, choose a K-representation W and a line $L \subseteq W$ with $H = \mathrm{Stab}_K(L)$.

Every finite-dimensional K-representation is a subquotient of a finite direct sum of tensor products of X and X^* [Mil12, Chapter VIII, Theorem 11.7]. Thus $W|_H$ belongs to $\mathcal{C}_X$, and the action of g on $W \otimes R$ agrees with λ_W: both are obtained from g by duals, tensor products, direct sums and passage to subquotients.

The line L is an H-subrepresentation of W. Naturality for its inclusion $i : L \hookrightarrow W$ gives

$$\begin{array}{ccc} L \otimes R & \xrightarrow{\lambda_L} & L \otimes R \\ {\scriptstyle i\otimes 1}\downarrow & & \downarrow{\scriptstyle i\otimes 1} \\ W \otimes R & \xrightarrow[\lambda_W]{} & W \otimes R. \end{array}$$

Since λ_L is invertible, $g(L \otimes R) = L \otimes R$. Hence $g \in H(R)$. Its action on X is λ_X, which determines all components on $\mathcal{C}_X$, so its action is exactly λ. This proves

$$G_X \simeq \underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X}).$$

These isomorphisms commute with restriction when $\mathcal{C}_X \subseteq \mathcal{C}_{X\oplus Y}$. The Hopf algebras $\mathcal{O}(G_X)$ are finitely generated as k-algebras and exhaust $\mathcal{O}(G)$, as proved

in Lecture 9. Passing to the compatible inverse limits gives

$$G \simeq \varprojlim_X G_X \simeq \varprojlim_X \underline{\mathrm{Aut}}^{\otimes}(\omega|_{\mathcal{C}_X}) \simeq \underline{\mathrm{Aut}}^{\otimes}(\omega).$$

All constructions are natural in R and respect the group operations. ∎

Functoriality

A group homomorphism $f : G \to G'$ gives restriction of representations

$$f^* : \mathrm{Rep}(G') \longrightarrow \mathrm{Rep}(G),$$

with the canonical identification $\omega_G f^* = \omega_{G'}$.

10.3 Corollary. *Group homomorphisms $G \to G'$ correspond to symmetric tensor functors $F : \mathrm{Rep}(G') \to \mathrm{Rep}(G)$ equipped with a specified tensor isomorphism $\alpha : \omega_G F \simeq \omega_{G'}$, up to isomorphisms respecting α.*

Proof. For each R, an element $g \in G(R)$ acts on every $F(V)_R$. Transport this action through α to obtain a tensor automorphism of $\omega_{G',R}$. Theorem 10.1 identifies it with a unique element of $G'(R)$. Naturality and multiplication give a group-scheme map $f : G \to G'$, and the same construction shows that α intertwines F with f^*. ∎

When k is algebraically closed, every symmetric tensor functor $F : \mathrm{Rep}(G') \to \mathrm{Rep}(G)$ is isomorphic to restriction along a group homomorphism $G \to G'$. Indeed, the two fiber functors $\omega_G F$ and $\omega_{G'}$ are tensor-isomorphic [Wib25, Corollary 1.2]; choosing such an isomorphism and applying the preceding corollary gives the homomorphism.

The general reconstruction problem

We next prove the following extension, in Lectures 11–13.

10.4 Theorem (Neutral Tannaka reconstruction)**.** *Let $\mathcal{C}$ be an essentially small symmetric tensor category over k, and let $\omega : \mathcal{C} \to \mathrm{Vec}_k$ be an exact faithful k-linear strong symmetric monoidal functor to finite-dimensional vector spaces. Then*

$$G = \underline{\mathrm{Aut}}^{\otimes}(\omega)$$

is an affine group scheme, and ω lifts to a symmetric tensor equivalence $\widetilde{\omega} : \mathcal{C} \simeq \mathrm{Rep}_k(G)$ whose composition with the forgetful functor is ω.

We first reconstruct a coalgebra from the abelian category and ω, then use the tensor product, symmetry and duals to recover multiplication, commutativity and the antipode.

References

[BEO20] Dave Benson, Pavel Etingof, and Victor Ostrik. New incompressible symmetric tensor categories in positive characteristic, 2020.

[CEO23] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. On Frobenius exact symmetric tensor categories. *Annals of Mathematics*, 197(3):1235–1279, 2023.

[CEO24] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. Incompressible tensor categories. *Advances in Mathematics*, 457:109935, 2024.

[CF24] Kevin Coulembier and Johannes Flake. Towards higher Frobenius functors for symmetric tensor categories, 2024. The course discussion follows the May 2024 version.

[Del02] Pierre Deligne. Catégories tensorielles. *Moscow Mathematical Journal*, 2(2):227–248, 2002.

[DM82] Pierre Deligne and James S. Milne. Tannakian categories. In *Hodge Cycles, Motives, and Shimura Varieties*, volume 900 of *Lecture Notes in Mathematics*, pages 101–228. Springer, 1982. Corrected version, 2022, with the original proposition numbering.

[EGNO15] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik. *Tensor categories*, volume 205 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2015.

[ENO05] Pavel Etingof, Dmitri Nikshych, and Viktor Ostrik. On fusion categories. *Annals of Mathematics*, 162(2):581–642, 2005.

[EO19] Pavel Etingof and Victor Ostrik. On semisimplification of tensor categories, 2019.

[Mil12] James S. Milne. Basic theory of affine group schemes, 2012. Available at www.jmilne.org/math/.

[Ost15] Victor Ostrik. On symmetric fusion categories in positive characteristic, 2015.

[Wat79] William. C. Waterhouse. *Introduction to Affine Group Schemes*. Springer Graduate texts in mathematics. Springer New York, NY, 1979.

[Wib25] Michael Wibmer. A remark on torsors under affine group schemes. *Transformation Groups*, 30:447–454, 2025.